

Illumination Chip — Engineering Pack + Quick Look

Seer Team,

We've packaged the Illumination Chip in an engineer-first format, with a clear separation between established physics and the frontier claim we're testing.

Included:

- Page 1 schematic (stack cross-section + rectifier chain)
- Quick Look protocol (60–90 min) for RF benches
- Replication checklist (energy balance & controls)
- Safety notes (creepage/clearance, EMC, interlocks, IEC refs)
- Example results tables

Bright lines:

Stone: piezo resonance, Q-mapping, static Casimir, DCE photon observation (as modulation-powered emission).

Frontier: net energy surplus beyond accounted inputs — treated strictly as a hypothesis, tested with open diagnostics.

If aligned, we'll proceed with preregistration of the test plan and share the zip (schematics/BOM/firmware/scripts) for replication.

Respect,
Gary
Illumination Engine Team

Quick Look Protocol (1 Page)

- 1 Resonance scan (VNA, capture f_{res} , Q).
- 2 Drive & mechanical lock (drive at f_{res} , log motional current/vibrometry).
- 3 DCE band check (spectrum analyzer around $f_{\text{res}}/2$, compare off/on, detune $\pm 1\%$).
- 4 Rectifier sanity (DC rise with drive, RF leakage < -60 dBm).
- 5 Mini energy balance (P_{in} absorbed vs P_{out} DC).

Replication Checklist

- Instrumentation calibrated (RF, DC, thermal).
- Faraday enclosure verified (no leakage).
- Controls: sham stack, off-res drive, amplitude scaling, fluid out.
- Energy: log P_{in} (absorbed) vs P_{out} (DC).
- Optional calorimetry agrees with electrical sums.

Safety Notes

- Working voltage: 350–400 V DC bus (hazardous).
- Clearance ≥ 3 mm; creepage ≥ 6 mm (better ≥ 8 mm).
- Grounding: single-point; isolation between RF/DC returns.
- Enclosure interlock removes drive when opened.
- Reference standards: IEC 62368-1, IEC 60664-1, IEC 61010-1.

Results Table (Example)

Run	f_drive (GHz)	P_fwd (mW)	P_refl (mW)	P_in (mW)	V_dc (V)	R_load (kΩ)	P_out (μW)	Notes
1	4.200	1.00	0.84	0.16	0.012	100	0.0014	Sham
2	4.200	1.00	0.70	0.30	0.031	100	0.0096	On■res, fluid